LLM Inference, Temperature, and Sampling

How a model turns context into generated text

Purpose

This short reference note is written for learning, semantic retrieval, and grounded question-answering experiments. It uses original explanatory content and is organized into coherent sections rather than unrelated filler.

1. Inference in a Language Model

Inference is the process of using a trained model to produce an output for a given input. In an autoregressive model, the system repeatedly predicts a distribution over the next token, selects a token according to a decoding strategy, appends it to the context, and continues until a stopping condition is reached. The prompt and any retrieved passages form part of the context available during generation.

2. Logits and Probabilities

The model produces scores often called logits for possible next tokens. A softmax transformation converts these scores into probabilities. The highest-probability token is not always selected: decoding settings may introduce controlled variation. Tokenization matters because the model selects token IDs, which are later decoded into readable text.

3. Greedy Decoding and Sampling

Greedy decoding selects the token with the highest probability at each step. It is relatively deterministic but can produce repetitive or generic output. Sampling draws from a probability distribution, allowing variation. Top-k sampling restricts candidates to the k most likely tokens; nucleus or top-p sampling keeps a set whose cumulative probability reaches a threshold. These methods trade predictability for diversity.

4. Temperature

Temperature adjusts the sharpness of the probability distribution before sampling. Lower temperatures make high-scoring tokens more dominant; higher temperatures flatten the distribution and increase variation. Temperature does not add knowledge to a model. For tasks that need stable, evidence-based answers, a low or moderate setting is often a sensible starting point, but the appropriate value depends on the model and application.

5. Context, Limits, and Stopping

The context window limits how much input and generated text can be handled in a single request. A request may also have a maximum output-token setting and stop sequences. If a prompt includes too much retrieved content, relevant information may be truncated or attention may be spread across irrelevant passages. Systems should budget tokens for instructions, evidence, user input, and the response.

6. Grounded Generation

For a RAG application, the prompt should clearly distinguish the user’s question from retrieved source passages. It can instruct the model to answer only when the evidence supports a claim, acknowledge missing information, and include source references. Such instructions reduce some unsupported responses but do not guarantee perfect grounding. The application should test supported questions and deliberately unsupported questions.

Key Takeaways

Inference combines a prompt, a model’s next-token distribution, and a decoding strategy. Temperature and sampling affect variation, while context limits and prompt structure affect how evidence can be used.

End of document • Use source passages as evidence and verify claims during evaluation.